

NUMERICAL MODELING AND SIMULATION FOR ACOUSTICS

A.A. 2025/2026

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HOMEWORK 2

Consider the linearized shallow water equations:

$$\begin{cases} \eta_t(x, t) + q_x(x, t) = 0 & (x, t) \in (0, L) \times (0, T), \\ q_t(x, t) + gH\eta_x(x, t) = 0 & (x, t) \in (0, L) \times (0, T), \end{cases} \quad (1)$$

where $\eta(x, t)$ is the free surface elevation, $q(x, t) = Hu(x, t)$ is the discharge with $u(x, t)$ the horizontal velocity, H is the mean depth, g is the gravity force, and $c = \sqrt{gH}$ is the wave speed. Together with (1) we consider the following periodic boundary conditions

$$\eta(0, t) = \eta(L, t), \quad q(0, t) = q(L, t).$$

1. Show that problem (1) is equivalent to

$$\eta_{tt}(x, t) = gH\eta_{xx}(x, t) \quad (x, t) \in (0, L) \times (0, T)$$

with suitable boundary conditions.

2. Show that the energy

$$E(t) = \int_0^L \left(\frac{1}{2} g \eta^2 + \frac{1}{2} \frac{q^2}{H} \right) dx$$

is conserved under periodic boundary conditions.

3. Write the weak formulation of problem (1) and the corresponding spectral element discretization. Show that the Galerkin discretization leads to the following system

$$M\dot{\mathbf{U}} + S\mathbf{U} = \mathbf{0}. \quad (2)$$

4. Implement the solver in MATLAB and verify your implementation by considering:

$$L = 1, \quad H = 1, \quad \text{and} \quad g = 9.81,$$

with initial conditions $\eta(x, 0) = e^{-50(x-0.5)^2}$ and $q(x, 0) = c\eta(x, 0)$, and final time $T = 0.5$. Compute and report the L^2 norm $\|\eta - \eta_h\|_{L^2(0, L)}$ for different choices of the discretization parameter h (mesh size) and polynomial degrees. Consider the exact solution $\eta(x, t) = \eta(x - ct, 0)$. Use a θ -method time integrator of the following form for (2):

$$M \frac{\mathbf{U}_{k+1} - \mathbf{U}_k}{\Delta t} + S[\theta \mathbf{U}_{k+1} + (1 - \theta) \mathbf{U}_k] = \mathbf{0}, \quad \theta \in [0, 1]. \quad (3)$$

As usual, the real positive parameter $\Delta t = t_{k+1} - t_k$, $k = 0, 1, \dots$, denotes the discretization step (here assumed to be constant), while the sub-index k indicates that the quantity under consideration refers to the time t_k . Suggestion: use $\theta = \frac{1}{2}$ in (3).

5. (*Optional.*) Study the effects on the solution of

- reflecting wall boundary conditions:

$$q = 0 \quad \text{at} \quad x = 0, L$$

- friction:

$$q_t(x, t) + gH\eta_x(x, t) = -\gamma q(x, t),$$

for a positive γ .